University of Minho
School of Engineering

Intelligent Systems Associate Laboratory



Dados e Aprendizagem Automática

Feature Engineering

Decision Tree Pruning and Hyperparameter Tuning

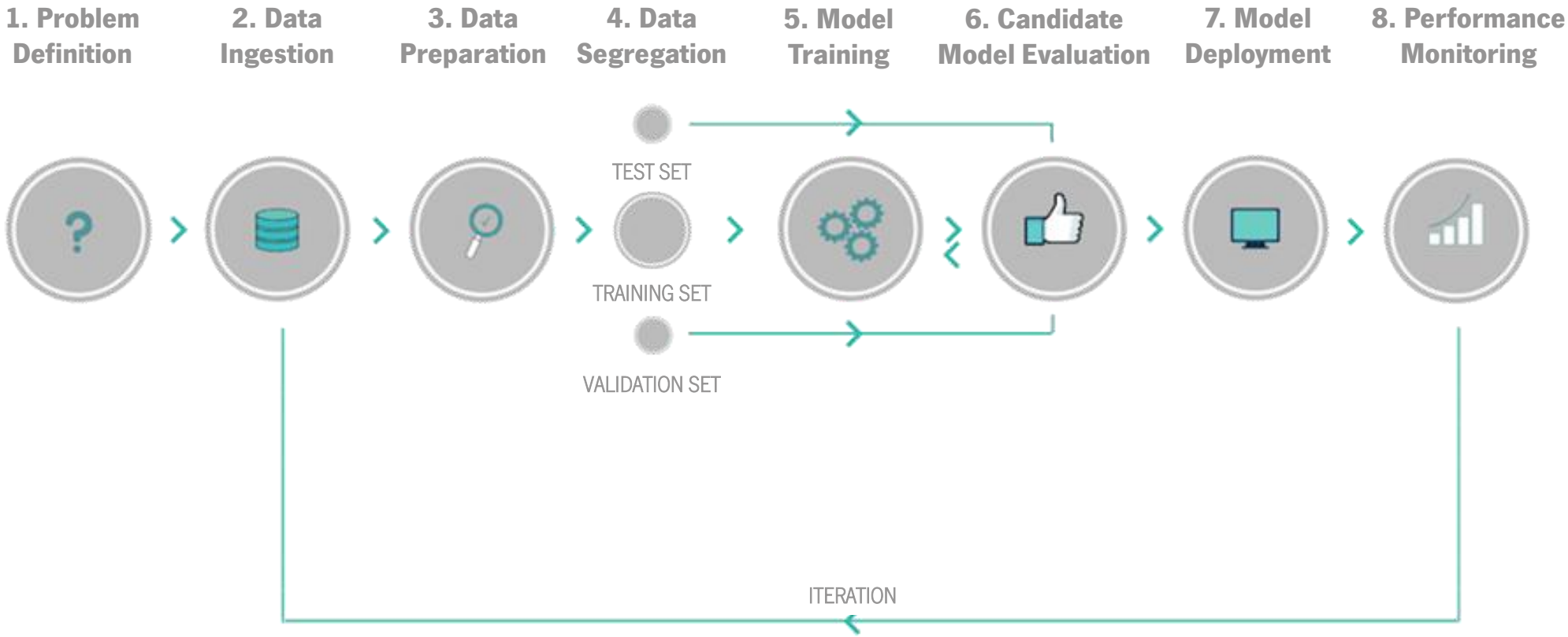
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Part V

Contents

- Feature Engineering
- Decision Tree
  - Hyperparameter Tuning with GridSearch
  - Pruning
- Hands On


1. Problem Definition
2. Data Ingestion
3. Data Preparation
4. Data Segregation
5. Model Training
6. Candidate Model Evaluation
7. Model Deployment
8. Performance Monitoring
TEST SET
TRAINING SET
VALIDATION SET
ITERATION

Feature Engineering

The Problem and the Data

Problem: development of a Machine Learning model that can **predict incidents** on the city of Guimarães' road network.

Dataset: information on the **road incidents** with 5000 entries and 13 features, including:

- *city_name*
- *magnitude_of_delay*
- *delay_in_seconds*
- *affected_roads*
- *record_date*
- *luminosity*
- *avg_temperature*
- *avg_atm_pressure*
- *avg_humidity*
- *avg_wind_speed*
- *avg_precipitation*
- *avg_rain*
- *incidents*

Exploratory Data Analysis


```
incid.columns
```

```
Index(['city_name', 'magnitude_of_delay', 'delay_in_seconds', 'affected_roads',
       'record_date', 'luminosity', 'avg_temperature', 'avg_atm_pressure',
       'avg_humidity', 'avg_wind_speed', 'avg_precipitation', 'avg_rain',
       'incidents'],
      dtype='object')
```

```
incid.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 13 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   city_name           5000 non-null   object 
 1   magnitude_of_delay  5000 non-null   object 
 2   delay_in_seconds    5000 non-null   int64  
 3   affected_roads      4915 non-null   object 
 4   record_date         5000 non-null   object 
 5   luminosity          5000 non-null   object 
 6   avg_temperature     5000 non-null   float64
 7   avg_atm_pressure    5000 non-null   float64
 8   avg_humidity        5000 non-null   float64
 9   avg_wind_speed      5000 non-null   float64
 10  avg_precipitation   5000 non-null   float64
 11  avg_rain            5000 non-null   object 
 12  incidents           2972 non-null   object 
dtypes: float64(5), int64(1), object(7)
memory usage: 507.9+ KB
```

Exploratory Data Analysis

7

```
incid.head()
```

	city_name	magnitude_of_delay	delay_in_seconds	affected_roads	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_preci
0	Guimaraes	UNDEFINED	0	,	2021-03-15 23:00	DARK	12.0	1013.0	70.0	1.0	
1	Guimaraes	UNDEFINED	385	N101,	2021-12-25 18:00	DARK	12.0	1007.0	91.0	1.0	
2	Guimaraes	UNDEFINED	69	,	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	
3	Guimaraes	MAJOR	2297	N101,R206,N105,N101,N101,N101,N101,N101,N101,N...	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	
4	Guimaraes	UNDEFINED	0	N101,N101,N101,N101,N101,	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	

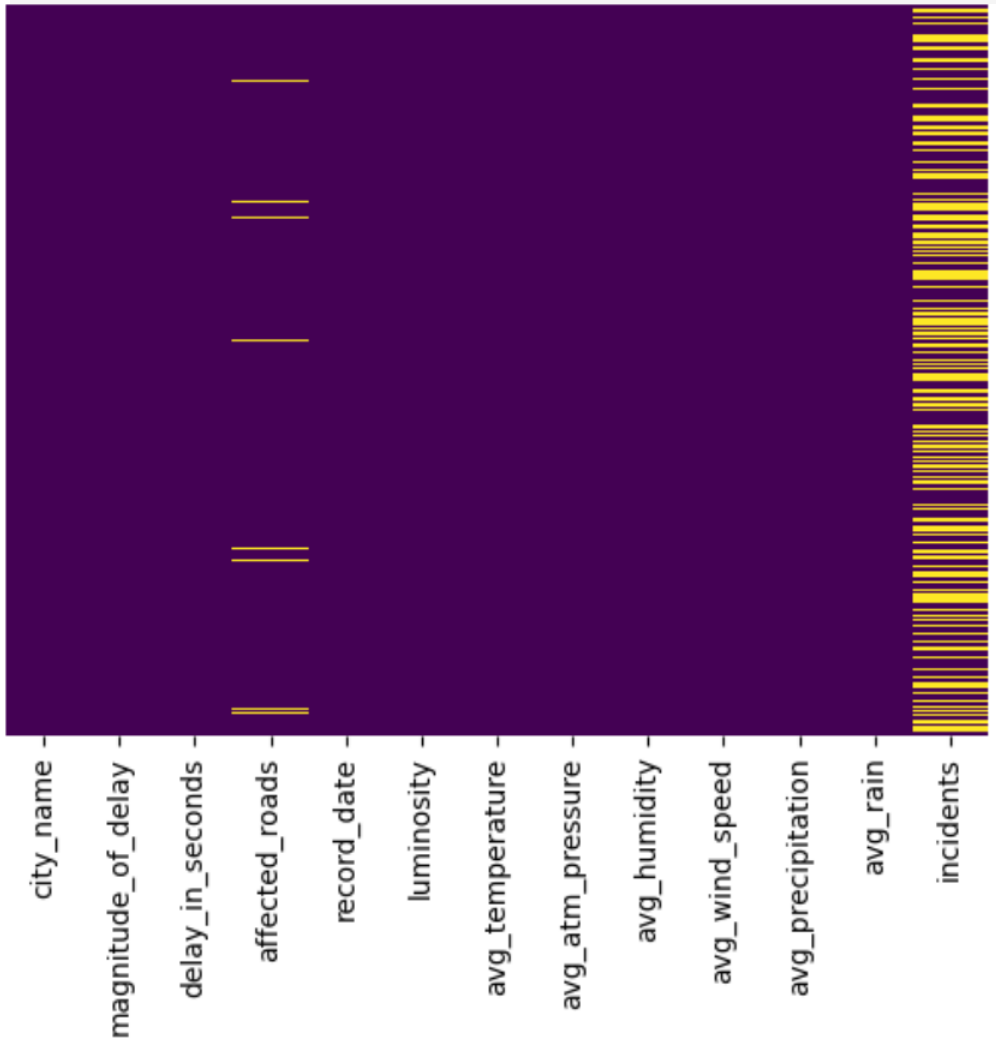
Feature Engineering

Handling missing data and data transformations

- Remove missing values, outliers, and unnecessary entries/features
- Check and impute null values
- Check imbalanced data
- Re-index and reformat the data

Dealing with Missing Values


```
sns.heatmap(incid.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```




```
incid.isnull().sum()
```

```
city_name                0
magnitude_of_delay       0
delay_in_seconds         0
affected_roads          85
record_date              0
luminosity               0
avg_temperature          0
avg_atm_pressure         0
avg_humidity             0
avg_wind_speed           0
avg_precipitation        0
avg_rain                 0
incidents             2028
dtype: int64
```

affected roads

```
incid['affected_roads'].head()
```

```
0                                                    ,
1                                                 N101,
2                                                    ,
3    N101,R206,N105,N101,N101,N101,N101,N101,N101,N...
4                                 N101,N101,N101,N101,N101,
Name: affected_roads, dtype: object
```

Dealing with Missing Values

10

```
incid[incid['affected_roads'].isnull()]
```

	city_name	magnitude_of_delay	delay_in_seconds	affected_roads	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_precipitation	avg_rain	incidents
29	Guimaraes	UNDEFINED	64	NaN	2021-01-22 09:00	LIGHT	8.0	1012.0	91.0	4.0	0.0	Sem Chuva	Medium
76	Guimaraes	UNDEFINED	223	NaN	2021-01-29 08:00	LIGHT	11.0	1022.0	92.0	1.0	0.0	Sem Chuva	High
79	Guimaraes	MAJOR	80	NaN	2021-12-24 21:00	DARK	11.0	1004.0	92.0	0.0	0.0	Sem Chuva	NaN
91	Guimaraes	UNDEFINED	52	NaN	2021-03-02 13:00	LIGHT	13.0	1024.0	78.0	2.0	0.0	Sem Chuva	Low
109	Guimaraes	UNDEFINED	139	NaN	2021-12-27 13:00	LIGHT	15.0	1014.0	88.0	5.0	0.0	Sem Chuva	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...

In order to handle missing values, we will first create copies of the data to experiment with different solutions:

```
incid1 = incid.copy()
incid2 = incid.copy()
```

Dealing with Missing Values

a) Remove the column

```
incid1.drop(['affected_roads'], axis = 1, inplace = True)
incid1.head()
```

	city_name	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_precipitation	avg_rain	incidents
0	Guimaraes	UNDEFINED	0	2021-03-15 23:00	DARK	12.0	1013.0	70.0	1.0	0.0	Sem Chuva	NaN
1	Guimaraes	UNDEFINED	385	2021-12-25 18:00	DARK	12.0	1007.0	91.0	1.0	0.0	Sem Chuva	NaN
2	Guimaraes	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	0.0	Sem Chuva	Low
3	Guimaraes	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	0.0	Sem Chuva	Very_High
4	Guimaraes	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	0.0	Sem Chuva	High

Dealing with Missing Values

b) Fill in the missing values with zero

```
incid2.fillna(0)
incid2.head()
```

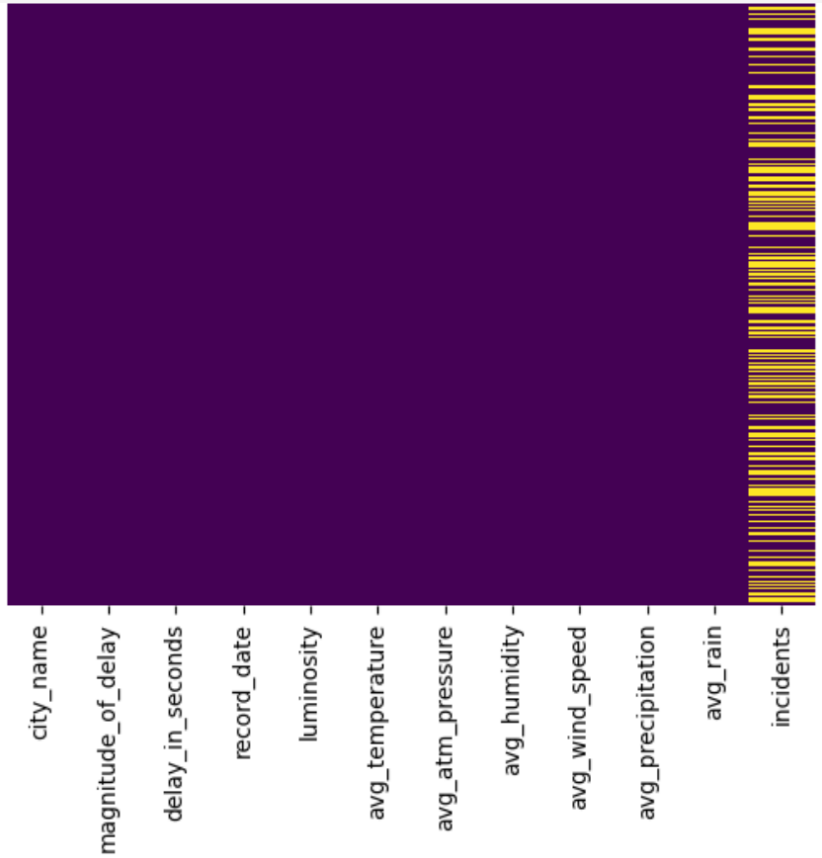
_delay	delay_in_seconds	affected_roads	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_precipitation	avg_rain	incidents
FINED	0	,	2021-03-15 23:00	DARK	12.0	1013.0	70.0	1.0	0.0	Sem Chuva	0
FINED	385	N101,	2021-12-25 18:00	DARK	12.0	1007.0	91.0	1.0	0.0	Sem Chuva	0
FINED	69	,	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	0.0	Sem Chuva	Low
MAJOR	2297	N101,R206,N105,N101,N101,N101,N101,N101,N101,N...	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	0.0	Sem Chuva	Very_High
FINED	0	N101,N101,N101,N101,N101,	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	0.0	Sem Chuva	High

Dealing with Missing Values

Since it does not bring added value to our goal, we will choose to remove the column.

```
incid.drop(['affected_roads'], axis = 1, inplace = True)
```

```
sns.heatmap(incid.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```




```
incid.isnull().sum()
```

```
city_name              0
magnitude_of_delay     0
delay_in_seconds       0
record_date            0
luminosity             0
avg_temperature        0
avg_atm_pressure       0
avg_humidity           0
avg_wind_speed         0
avg_precipitation      0
avg_rain               0
incidents           2028
dtype: int64
```

```
incid.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   city_name           5000 non-null   object
 1   magnitude_of_delay  5000 non-null   object
 2   delay_in_seconds    5000 non-null   int64
 3   record_date         5000 non-null   object
 4   luminosity          5000 non-null   object
 5   avg_temperature     5000 non-null   float64
 6   avg_atm_pressure    5000 non-null   float64
 7   avg_humidity        5000 non-null   float64
 8   avg_wind_speed      5000 non-null   float64
 9   avg_precipitation   5000 non-null   float64
 10  avg_rain            5000 non-null   object
 11  incidents           2972 non-null   object
dtypes: float64(5), int64(1), object(6)
memory usage: 468.9+ KB
```

Dealing with Unique Values


```
incid.nunique()
```

```
city_name              1
magnitude_of_delay     3
delay_in_seconds    1186
record_date         5000
luminosity             3
avg_temperature       35
avg_atm_pressure      36
avg_humidity          83
avg_wind_speed        11
avg_precipitation      1
avg_rain               4
incidents              4
dtype: int64
```

The *city_name* and *avg_precipitation* features have only one value.

avg_precipitation

```
incid['avg_precipitation'].nunique()
```

```
1
```

```
incid['avg_precipitation'].describe()
```

```
count    5000.0
mean        0.0
std         0.0
min         0.0
25%         0.0
50%         0.0
75%         0.0
max         0.0
Name: avg_precipitation, dtype: float64
```

Dealing with Unique Values

As 0 is the unique value for *avg_precipitation* and all entries have this value, we will remove this feature.

```
incid.drop(['avg_precipitation'], axis = 1, inplace = True)
incid.head()
```

	city_name	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
0	Guimaraes	UNDEFINED	0	2021-03-15 23:00	DARK	12.0	1013.0	70.0	1.0	Sem Chuva	NaN
1	Guimaraes	UNDEFINED	385	2021-12-25 18:00	DARK	12.0	1007.0	91.0	1.0	Sem Chuva	NaN
2	Guimaraes	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	Low
3	Guimaraes	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	Very_High
4	Guimaraes	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	High

Dealing with Unique Values and Handling Categoric Data

Although some models accept categorical data, we will explore how it can be transformed.

city_name

```
incid['city_name'].head()
```

```
0    Guimaraes
1    Guimaraes
2    Guimaraes
3    Guimaraes
4    Guimaraes
Name: city_name, dtype: object
```

The unique value of the *city_name* feature is “Guimarães” – we can drop this feature as well.

```
incid.drop('city_name',axis=1,inplace=True)
incid.dropna(inplace=True)
```

Handling Categoric Data

incidents

```
print(incid['incidents'].value_counts())
```

```
incidents
High         1073
Low           718
Very_High     603
Medium        578
Name: count, dtype: int64
```

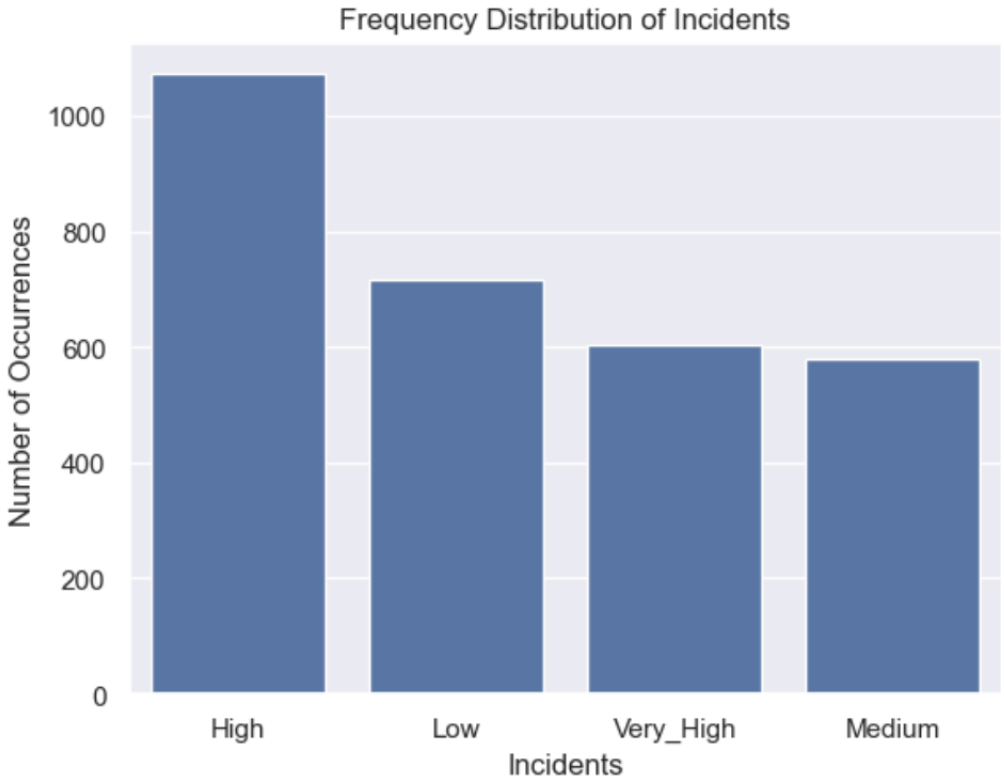
```
print(incid['incidents'].value_counts().count())
```

```
4
```

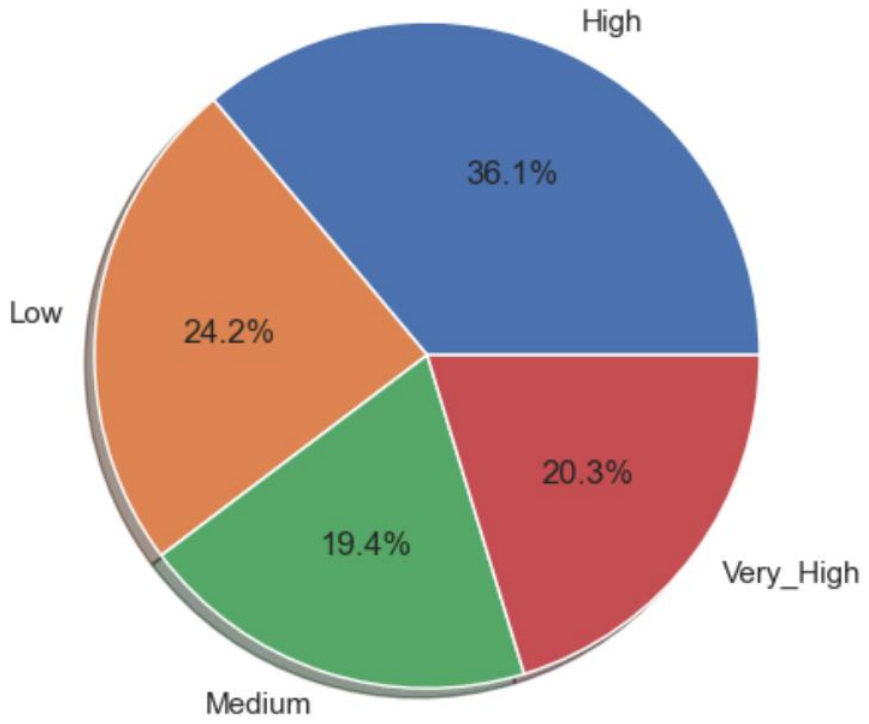
We will use visualizations to see how the values of feature *incidents* are distributed.

Handling Categoric Data


```
incidents_count = incid['incidents'].value_counts()
sns.set(style="darkgrid")
sns.barplot(x=incidents_count.index, y=incidents_count.values)
plt.title('Frequency Distribution of Incidents')
plt.ylabel('Number of Occurrences', fontsize=12)
plt.xlabel('Incidents', fontsize=12)
plt.show()
```




```
labels = incid['incidents'].astype('category').cat.categories.tolist()
counts = incid['incidents'].value_counts()
sizes = [counts[var_cat] for var_cat in labels]
fig1, ax1 = plt.subplots()
ax1.pie(sizes, labels=labels, autopct='%1.1f%%', shadow=True)
ax1.axis('equal')
plt.show()
```



Handling Categoric Data

a) Replace values

```
incid3=incid.copy()
incid3.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	Low
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	Very_High
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	High
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	Medium
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	Low

Label association: *None* - 0, *Low* - 1, *Medium* - 2, *High* - 3, *Very_High* - 4

```
replace_map = {'incidents': {'None': 0, 'Low': 1, 'Medium': 2, 'High': 3, 'Very_High': 4}}
```

Handling Categoric Data

Create a replacement map using a different method:

```
labels = incid3['incidents'].astype('category').cat.categories.tolist()
replace_map_comp = {'incidents' : {k: v for k,v in zip(labels, list(range(1, len(labels)+1)))}}
print(replace_map_comp)
```

```
{'incidents': {'High': 1, 'Low': 2, 'Medium': 3, 'Very_High': 4}}
```

```
incid3.replace(replace_map_comp, inplace=True)
incid3.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	2
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	4
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	1
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	3
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	2

```
print(incid3['incidents'].dtypes)
```

```
int64
```

Handling Categoric Data

b) Label encoding

As in the previous examples, each string is assigned a number. Rather than replacing the values under the *incidents* column, we will create a new column for each created label. To complete the process, this must be replicated for each label, after which the *incidents* column can be removed.

```
incid4=incid.copy()
```

```
print(incid4.dtypes)
```

```
magnitude_of_delay     object
delay_in_seconds        int64
record_date            object
luminosity             object
avg_temperature       float64
avg_atm_pressure      float64
avg_humidity          float64
avg_wind_speed        float64
avg_rain               object
incidents              object
dtype: object
```

```
incid4['None'] = np.where(incid4['incidents'].str.contains('None'), 1, 0)
incid4.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	None
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	Low	0
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	Very_High	0
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	High	0
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	Medium	0
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	Low	0

Handling Categoric Data

b) Label encoding using `LabelEncoder()`

```
incid5=incid.copy()
incid6=incid.copy()

from sklearn.preprocessing import LabelEncoder

le_make = LabelEncoder()
incid5['incidents_code'] = le_make.fit_transform(incid6['incidents'])

incid5.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	incidents_code
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	Low	1
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	Very_High	3
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	High	0
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	Medium	2
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	Low	1

A new column, *incidents_code*, has been created, to which labels have been assigned for *incidents* feature. The numeric values have been assigned randomly, since crescent order does not apply to the meaning of the qualifying words.

Handling Categoric Data

c) One-Hot Encoding

This alternative generates a matrix with bits containing information about each label.

```
incid7 = incid.copy()

from sklearn.preprocessing import LabelBinarizer

lb = LabelBinarizer()
lb_results = lb.fit_transform(incid7['incidents'])
lb_results_df = pd.DataFrame(lb_results, columns=lb.classes_)

lb_results_df.head()
```

	High	Low	Medium	Very_High
0	0	1	0	0
1	0	0	0	1
2	1	0	0	0
3	0	0	1	0
4	0	1	0	0

Handling Categoric Data


```
result_df = pd.concat([incid7, lb_results_df], axis=1)
result_df.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	High	Low	Medium	Very_High
2	UNDEFINED	69.0	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	Low	1.0	0.0	0.0	0.0
3	MAJOR	2297.0	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	Very_High	0.0	0.0	1.0	0.0
4	UNDEFINED	0.0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	High	0.0	1.0	0.0	0.0
5	UNDEFINED	0.0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	Medium	1.0	0.0	0.0	0.0
6	UNDEFINED	0.0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	Low	0.0	0.0	0.0	1.0

Handling Categoric Data

d) Binary Encoding

Similar to the previous technique, this one creates a matrix of the values' states, but this time with binary values. See the comparison between the three techniques below:

Level	"Decimal encoding"	Binary encoding	One-Hot encoding
None	0	000	000001
Low	1	001	000010
Medium	2	010	000100
High	3	011	001000
Very_High	4	100	010000

This technique requires the `category_encoders` package to be installed.

```
pip install category_encoders
```

Handling Categoric Data


```
incid8 = incid.copy()

import category_encoders as ce

bencoder = ce.BinaryEncoder(cols=['incidents'])
df_binary = bencoder.fit_transform(incid8)

df_binary.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents_0	incidents_1	incidents_2
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	0	0	1
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	0	1	0
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	0	1	1
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	1	0	0
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	0	0	1

Handling Categoric Data

e) Backward Difference Encoding

The values are normalized within the range of -1 to 1.

```
incid9 = incid.copy()

bde = ce.BackwardDifferenceEncoder(cols=['incidents'])
df_bd = bde.fit_transform(incid9)

df_bd.head()
```

	intercept	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents_0	incidents_1	incidents_2
2	1	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	-0.75	-0.5	-0.25
3	1	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	0.25	-0.5	-0.25
4	1	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	0.25	0.5	-0.25
5	1	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	0.25	0.5	0.75
6	1	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	-0.75	-0.5	-0.25

Handling Categoric Data

f) Factorizing

This technique involves encoding the object as an enumerated type or categorical variable.

```
incid10 = incid.copy()
```

```
incid10['incidents'] = pd.factorize(incid10['incidents'])[0] + 1
incid10.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
2	UNDEFINED	69	2021-03-12 15:00	LIGHT	14.0	1025.0	64.0	0.0	Sem Chuva	1
3	MAJOR	2297	2021-09-29 09:00	LIGHT	15.0	1028.0	75.0	1.0	Sem Chuva	2
4	UNDEFINED	0	2021-06-13 11:00	LIGHT	27.0	1020.0	52.0	1.0	Sem Chuva	3
5	UNDEFINED	0	2021-12-07 23:00	DARK	9.0	1015.0	94.0	0.0	Sem Chuva	4
6	UNDEFINED	0	2021-12-05 05:00	DARK	8.0	1026.0	87.0	1.0	Sem Chuva	1

We will use the factorizing technique to continue.

```
incid['incidents'] = pd.factorize(incid['incidents'])[0] + 1
```

Handling Categoric Data

magnitude_of_delay*, *luminosity* and *avg_rain

For now, we will factor out these features.

```
incid['magnitude_of_delay'] = pd.factorize(incid['magnitude_of_delay'])[0] + 1
incid['luminosity'] = pd.factorize(incid['luminosity'])[0] + 1
incid['avg_rain'] = pd.factorize(incid['avg_rain'])[0] + 1

incid.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
2	1	69	2021-03-12 15:00	1	14.0	1025.0	64.0	0.0	1	1
3	2	2297	2021-09-29 09:00	1	15.0	1028.0	75.0	1.0	1	2
4	1	0	2021-06-13 11:00	1	27.0	1020.0	52.0	1.0	1	3
5	1	0	2021-12-07 23:00	2	9.0	1015.0	94.0	0.0	1	4
6	1	0	2021-12-05 05:00	2	8.0	1026.0	87.0	1.0	1	1

Handling Dates

record_date

There are several techniques for handling dates. We will explore just a few of these. The `datetime` properties and methods can be consulted here.

```
incid11 = incid.copy()
incid11['record_date'].head()
```

```
2    2021-03-12 15:00
3    2021-09-29 09:00
4    2021-06-13 11:00
5    2021-12-07 23:00
6    2021-12-05 05:00
Name: record_date, dtype: object
```

We will convert the dates from `object` to `datetime`, specifying the desired format.

```
incid11['record_date'] = pd.to_datetime(incid11['record_date'], format = '%Y-%m-%d %H:%M', errors='coerce')
assert incid11['record_date'].isnull().sum() == 0, 'missing record date'
incid11['record_date'].head()
```

```
2   2021-03-12 15:00:00
3   2021-09-29 09:00:00
4   2021-06-13 11:00:00
5   2021-12-07 23:00:00
6   2021-12-05 05:00:00
Name: record_date, dtype: datetime64[ns]
```

Handling Dates

We can extract parts of the date to create new columns.

```
incid11['record_date_year'] = incid11['record_date'].dt.year
incid11['record_date_month'] = incid11['record_date'].dt.month
incid11['record_date_day'] = incid11['record_date'].dt.day
incid11['record_date_hour'] = incid11['record_date'].dt.hour
incid11['record_date_minute'] = incid11['record_date'].dt.minute
```

```
incid11.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	record_date_year	record_date_month	rec
2	1	69	2021-03-12 15:00:00	1	14.0	1025.0	64.0	0.0	1	1	2021	3	
3	2	2297	2021-09-29 09:00:00	1	15.0	1028.0	75.0	1.0	1	2	2021	9	
4	1	0	2021-06-13 11:00:00	1	27.0	1020.0	52.0	1.0	1	3	2021	6	
5	1	0	2021-12-07 23:00:00	2	9.0	1015.0	94.0	0.0	1	4	2021	12	
6	1	0	2021-12-05 05:00:00	2	8.0	1026.0	87.0	1.0	1	1	2021	12	

Handling Dates


```
incid11.nunique()
```

```
magnitude_of_delay       3
delay_in_seconds      1167
record_date           2972
luminosity               3
avg_temperature         34
avg_atm_pressure        34
avg_humidity            80
avg_wind_speed          11
avg_rain                 4
incidents                4
record_date_year         1
record_date_month       11
record_date_day         31
record_date_hour        24
record_date_minute       1
dtype: int64
```

Since both the *year* and the *minute* only have one value, we will remove them.

```
incid11.drop('record_date_year',axis=1,inplace=True)
incid11.drop('record_date_minute',axis=1,inplace=True)
incid11.drop('record_date',axis=1,inplace=True)
incid11.dropna(inplace=True)
```

Handling Dates


```
incid11.head()
```

	magnitude_of_delay	delay_in_seconds	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	record_date_month	record_date_day	record_date_hour
2	1	69	1	14.0	1025.0	64.0	0.0	1	1	3	12	15
3	2	2297	1	15.0	1028.0	75.0	1.0	1	2	9	29	9
4	1	0	1	27.0	1020.0	52.0	1.0	1	3	6	13	11
5	1	0	2	9.0	1015.0	94.0	0.0	1	4	12	7	23
6	1	0	2	8.0	1026.0	87.0	1.0	1	1	12	5	5

Other methods to deal with dates:

```
incid12 = incid.copy()
```

```
incid12['record_date'] = pd.to_datetime(incid12['record_date'])
```

```
incid12.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 2972 entries, 2 to 4995
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   magnitude_of_delay  2972 non-null   int64
 1   delay_in_seconds    2972 non-null   int64
 2   record_date         2972 non-null   datetime64[ns]
 3   luminosity          2972 non-null   int64
 4   avg_temperature     2972 non-null   float64
 5   avg_atm_pressure    2972 non-null   float64
 6   avg_humidity        2972 non-null   float64
 7   avg_wind_speed      2972 non-null   float64
 8   avg_rain            2972 non-null   int64
 9   incidents           2972 non-null   int64
dtypes: datetime64[ns](1), float64(4), int64(5)
memory usage: 319.9 KB
```

```
incid12['record_date'].head()
```

```
2   2021-03-12 15:00:00
3   2021-09-29 09:00:00
4   2021-06-13 11:00:00
5   2021-12-07 23:00:00
6   2021-12-05 05:00:00
Name: record_date, dtype: datetime64[ns]
```

Handling Dates


```
import datetime
today = datetime.datetime.today()
today
```

```
datetime.datetime(2024, 10, 14, 17, 50, 16, 977144)
```

```
today - incid12['record_date']
```

```
2      1312 days 02:50:16.977144
3      1111 days 08:50:16.977144
4      1219 days 06:50:16.977144
5      1041 days 18:50:16.977144
6      1044 days 12:50:16.977144
                  ...
4991   1115 days 23:50:16.977144
4992   1254 days 06:50:16.977144
4993   1209 days 01:50:16.977144
4994   1208 days 19:50:16.977144
4995   1280 days 17:50:16.977144
Name: record_date, Length: 2972, dtype: timedelta64[ns]
```

```
(today - incid12['record_date']).dt.days
```

```
2       1312
3       1111
4       1219
5       1041
6       1044
        ...
4991    1115
4992    1254
4993    1209
4994    1208
4995    1280
Name: record_date, Length: 2972, dtype: int64
```

```
incid12['day'] = incid12['record_date'].dt.day
incid12['month'] = incid12['record_date'].dt.month
incid12['hour'] = incid12['record_date'].dt.hour
incid12['time'] = incid12['record_date'].dt.time
```

```
incid12.head()
```

	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents	day	month	hour	time
2	1	69	2021-03-12 15:00:00	1	14.0	1025.0	64.0	0.0	1	1	12	3	15	15:00:00
3	2	2297	2021-09-29 09:00:00	1	15.0	1028.0	75.0	1.0	1	2	29	9	9	09:00:00
4	1	0	2021-06-13 11:00:00	1	27.0	1020.0	52.0	1.0	1	3	13	6	11	11:00:00
5	1	0	2021-12-07 23:00:00	2	9.0	1015.0	94.0	0.0	1	4	7	12	23	23:00:00

Handling Dates


```
incid['record_date'] = pd.to_datetime(incid['record_date'], format = '%Y-%m-%d %H:%M', errors='coerce')
```

```
incid.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 2972 entries, 2 to 4995
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   magnitude_of_delay  2972 non-null   int64
 1   delay_in_seconds    2972 non-null   int64
 2   record_date         2972 non-null   datetime64[ns]
 3   luminosity          2972 non-null   int64
 4   avg_temperature     2972 non-null   float64
 5   avg_atm_pressure    2972 non-null   float64
 6   avg_humidity        2972 non-null   float64
 7   avg_wind_speed      2972 non-null   float64
 8   avg_rain            2972 non-null   int64
 9   incidents           2972 non-null   int64
dtypes: datetime64[ns](1), float64(4), int64(5)
memory usage: 319.9 KB
```

```
incid.head()
```

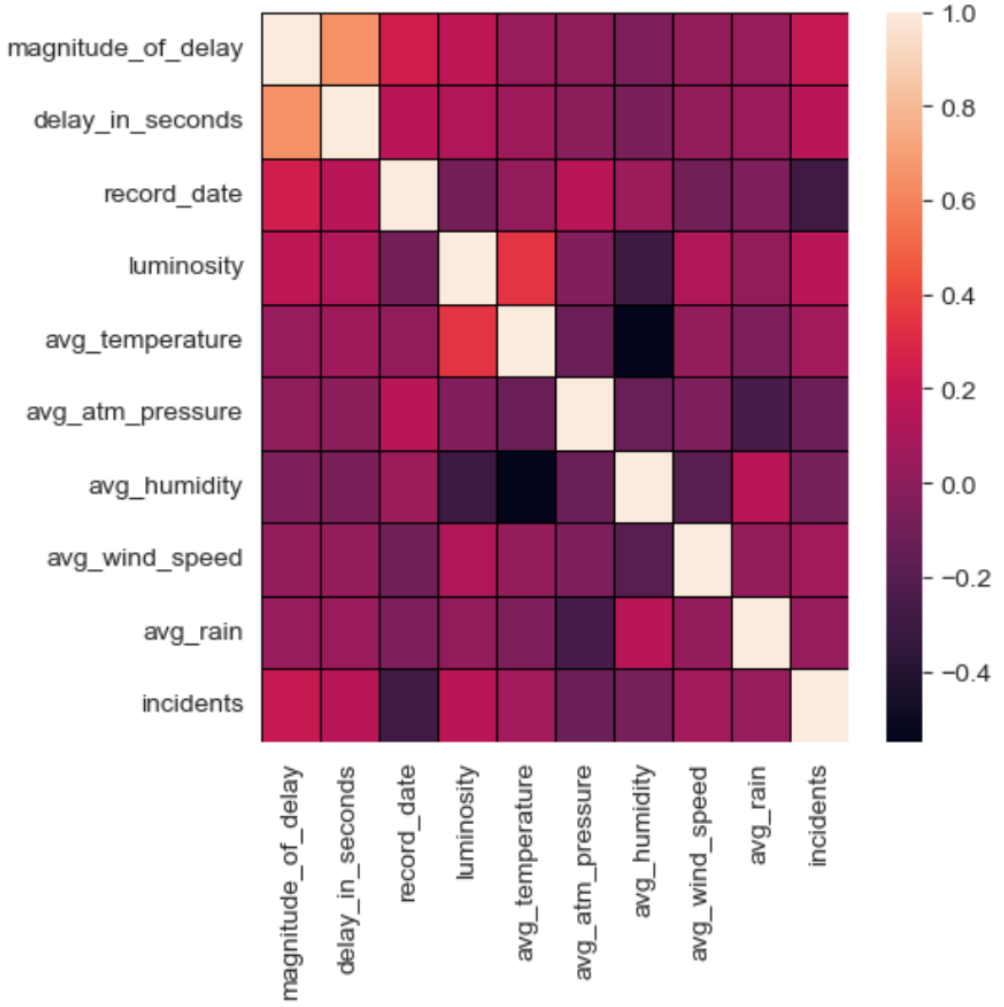
	magnitude_of_delay	delay_in_seconds	record_date	luminosity	avg_temperature	avg_atm_pressure	avg_humidity	avg_wind_speed	avg_rain	incidents
2	1	69	2021-03-12 15:00:00	1	14.0	1025.0	64.0	0.0	1	1
3	2	2297	2021-09-29 09:00:00	1	15.0	1028.0	75.0	1.0	1	2
4	1	0	2021-06-13 11:00:00	1	27.0	1020.0	52.0	1.0	1	3
5	1	0	2021-12-07 23:00:00	2	9.0	1015.0	94.0	0.0	1	4
6	1	0	2021-12-05 05:00:00	2	8.0	1026.0	87.0	1.0	1	1

Decision Tree

EDA

Before approaching the model, we need to analyze the data and search for relations between features.

```
fig = plt.figure(figsize = (5,5))
incidents_corr = incid.corr( method = 'pearson')
sns.heatmap(incidents_corr, linecolor='black', linewidths=0.5)
```



The highlighted relations are:

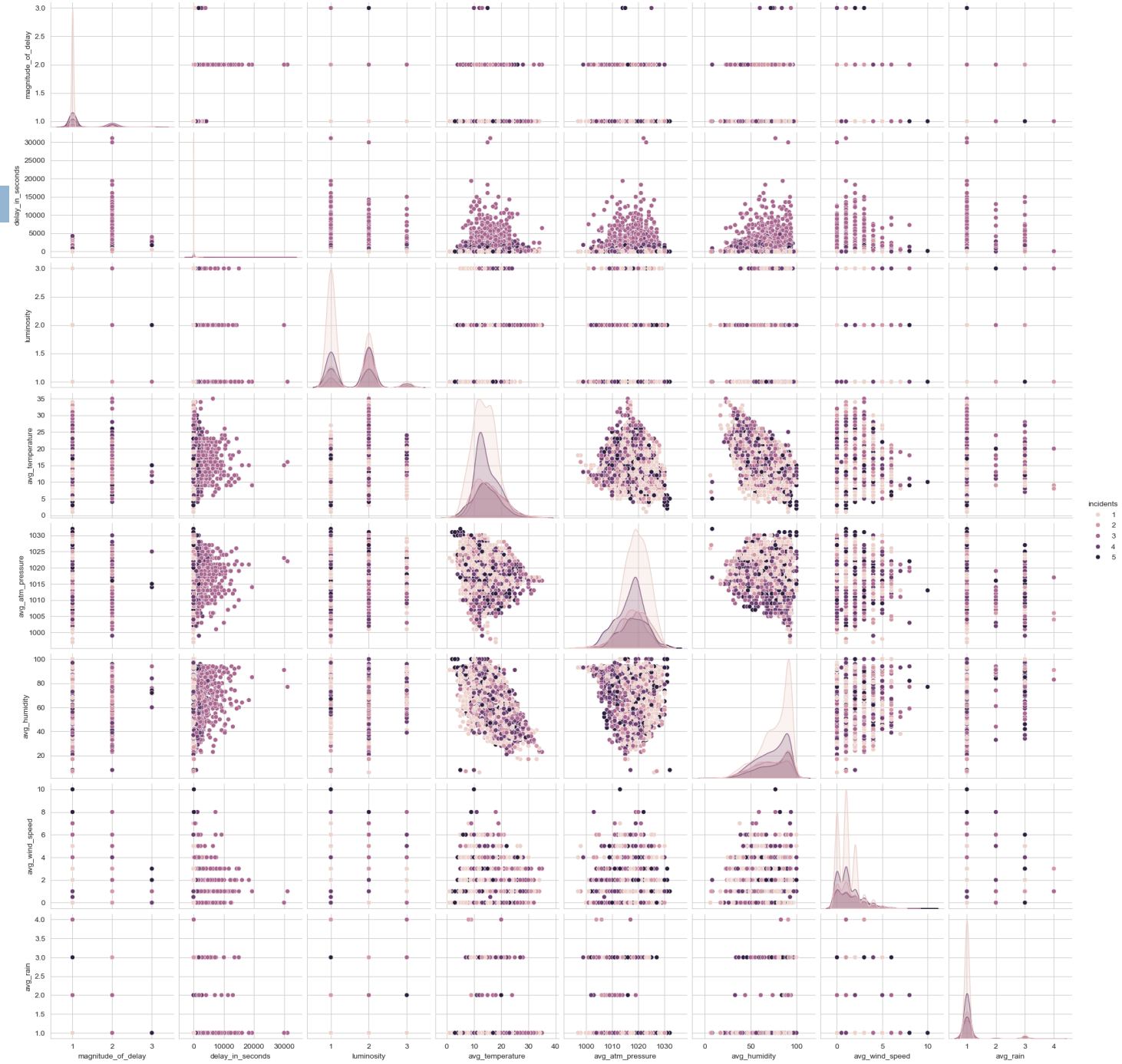
- *magnitude_of_delay* and *delay_in_seconds*
- *magnitude_of_delay* and *record_date*
- *avg_humidity* and *luminosity*

EDA


```
sns_plot = sns.pairplot(incid, hue="incidents")
sns_plot.savefig('pp.png')
```

Analyzing the relationship between all the features is **difficult**.

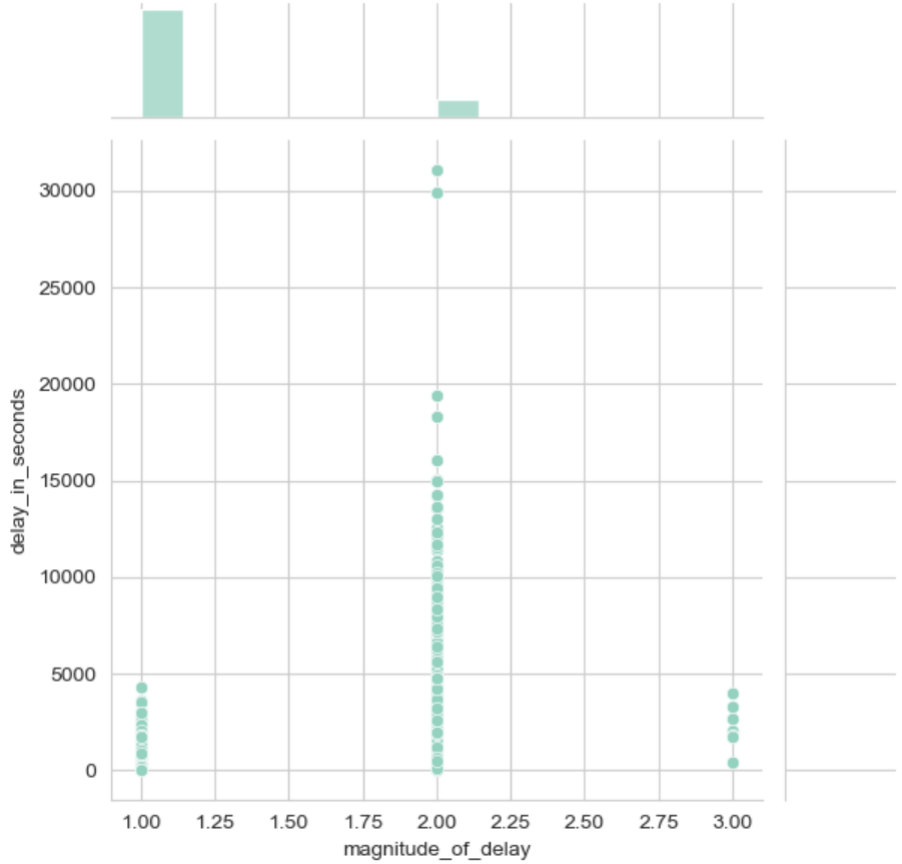
Create **joint plots** showing the relationships between the features.



EDA

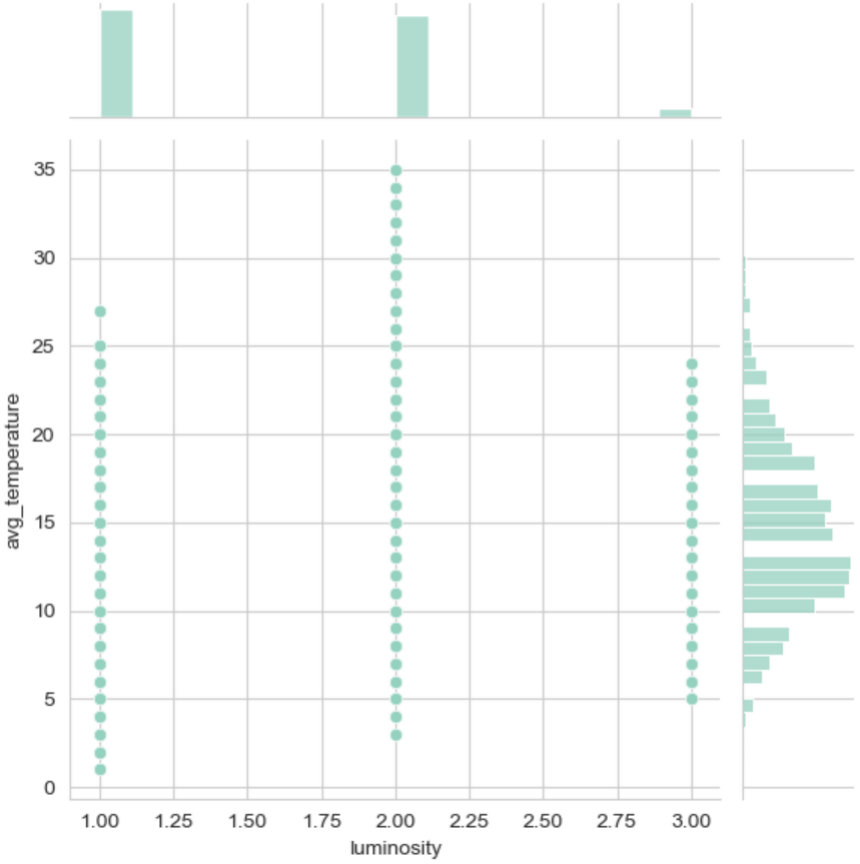
magnitude_of_delay vs ***delay_in_seconds***

```
sns.set_palette("GnBu_d")
sns.set_style('whitegrid')
sns.jointplot(x='magnitude_of_delay', y='delay_in_seconds', data=incid)
```



luminosity vs ***avg_temperature***

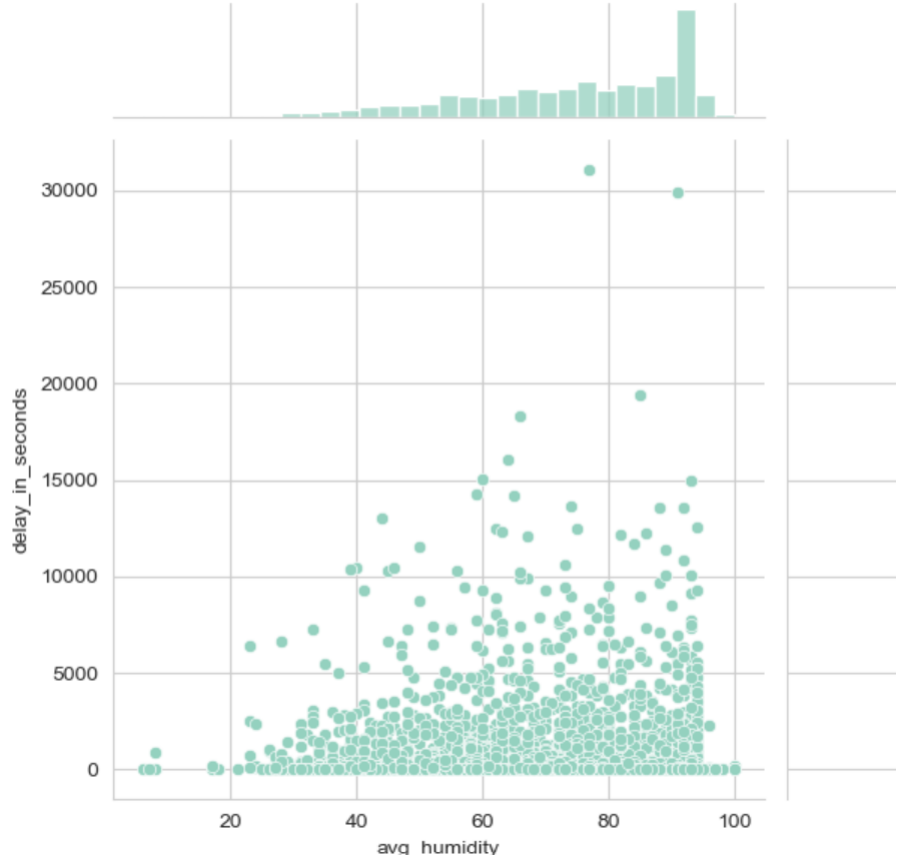
```
sns.jointplot(x='luminosity', y='avg_temperature', data=incid)
```



EDA

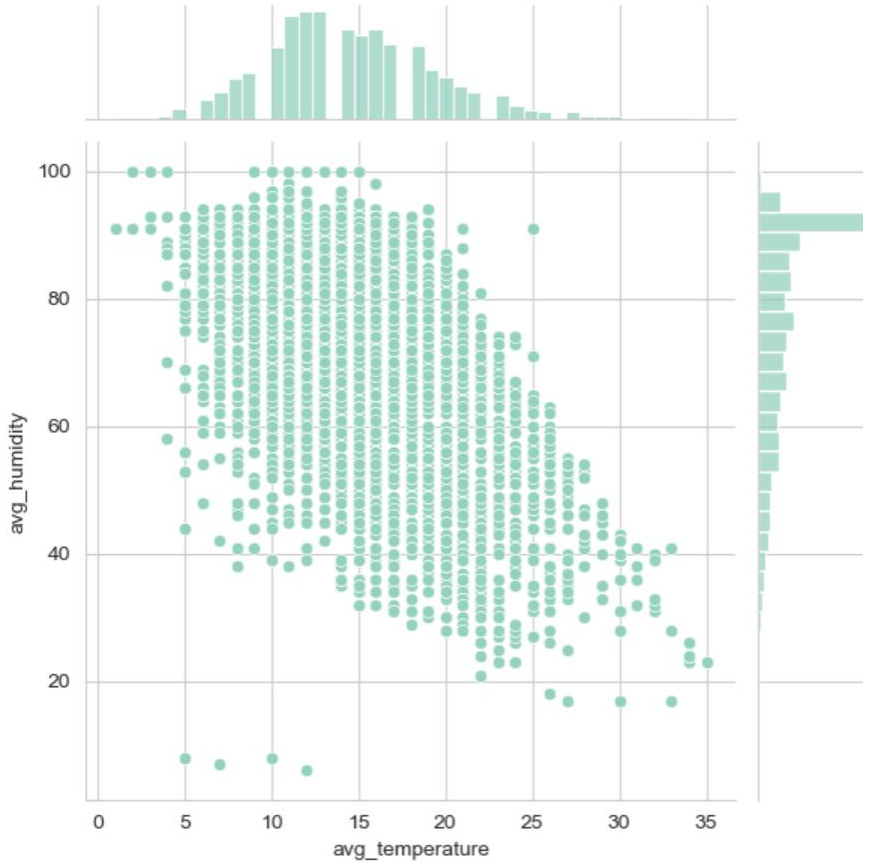
avg_humidity vs ***delay_in_seconds***

```
sns.jointplot(x='avg_humidity', y='delay_in_seconds', data=incid)
```



avg_temperature vs ***avg_humidity***

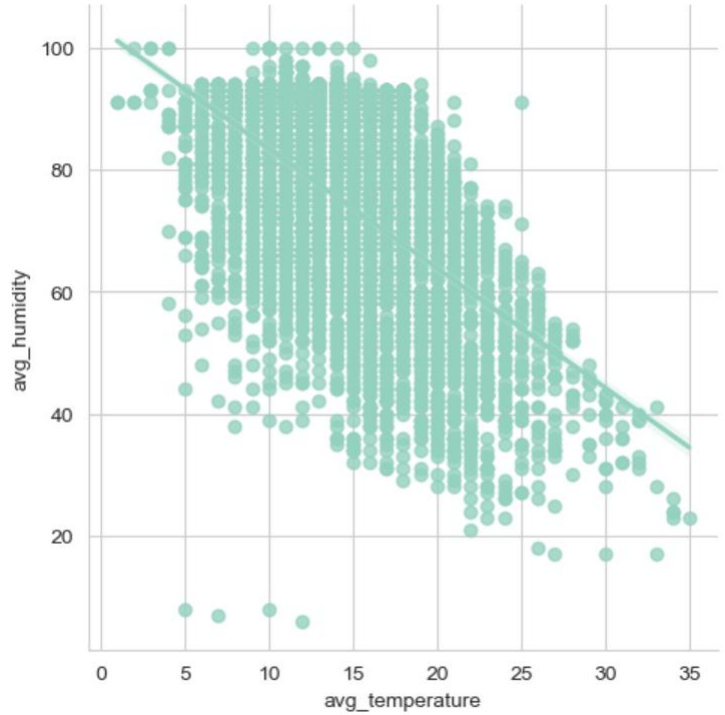
```
sns.jointplot(x='avg_temperature', y='avg_humidity', data=incid)
```



EDA

It appears that there is a relationship between *avg_temperature* and *avg_humidity*.

```
sns.lmplot(x='avg_temperature', y='avg_humidity', data=incid)
```



Decision Tree Classifier

Since the target is the *incidents* feature, we will implement a DT Classifier model.
The *record_date* feature will not be considered, as it is irrelevant to the context.

```
from sklearn.model_selection import train_test_split
```

```
X = incid.drop(['record_date', 'incidents'], axis=1)
y = incid['incidents']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, random_state=2022)
```

Decision Tree Classifier


```
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
```

```
sklearn.tree.DecisionTreeClassifier(*, criterion='gini', splitter='best', max_depth=None, min_samples_split=2,
                                    min_samples_leaf=1, min_weight_fraction_leaf=0.0, max_features=None, random_state=None, max_leaf_nodes=None,
                                    min_impurity_decrease=0.0, class_weight=None, ccp_alpha=0.0, monotonic_cst=None)
```

```
dt_model = DecisionTreeClassifier(random_state=2022)
```

```
dt_model.fit(X_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=2022)
```

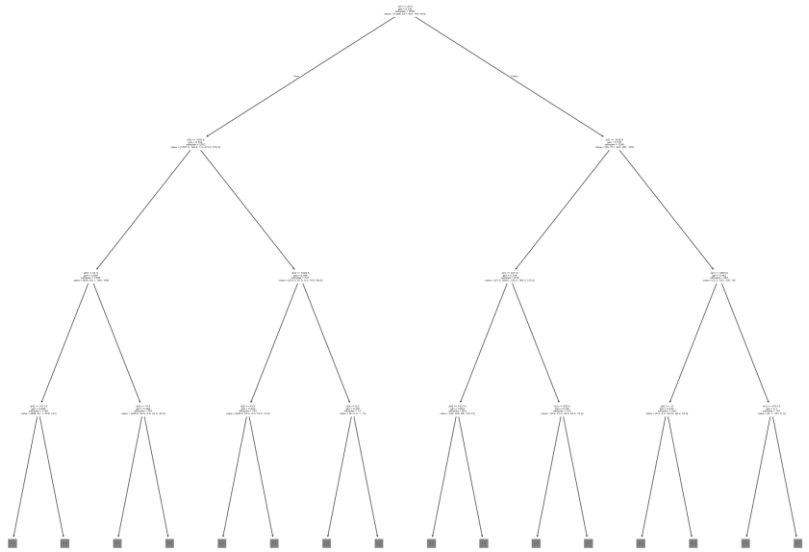
Plot the resultant tree and save it as a figure or export it as text.

```
fig = plt.figure(figsize=(25, 20))
tree.plot_tree(dt_model, max_depth = 3)
plt.show()

fig.savefig("dt_plot.png")
```

```
text_representation = tree.export_text(dt_model)
print(text_representation)

with open("dt_text.log", "w") as fout:
    fout.write(text_representation)
```



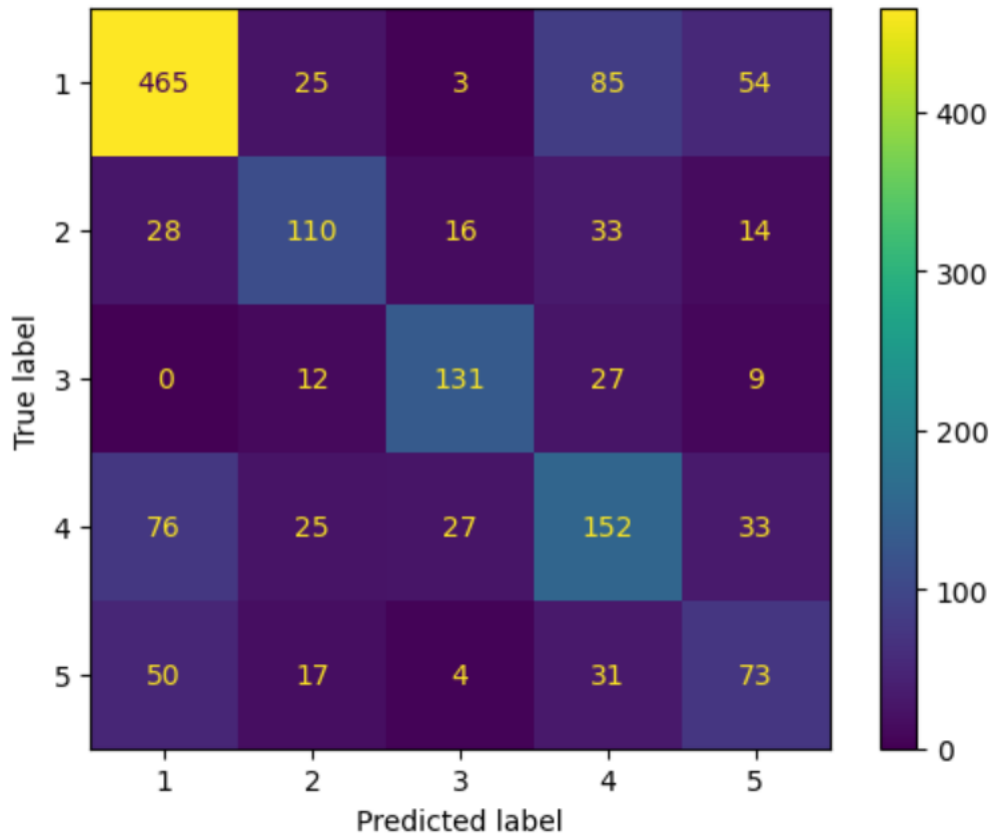
Decision Tree Classifier

Evaluate the model:

```
dt_predictions = dt_model.predict(X_test)
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
print(classification_report(y_test, dt_predictions))
```

	precision	recall	f1-score	support
1	0.75	0.74	0.74	632
2	0.58	0.55	0.56	201
3	0.72	0.73	0.73	179
4	0.46	0.49	0.47	313
5	0.40	0.42	0.41	175
accuracy			0.62	1500
macro avg	0.58	0.58	0.58	1500
weighted avg	0.62	0.62	0.62	1500

```
ConfusionMatrixDisplay.from_predictions(y_test, dt_predictions)
plt.show()
```



Parameters and Hyperparameters

Model parameter: an internal configuration variable of the model whose value is estimated from training data; in other words, the value is not set manually. Some examples are:

- weights in Artificial Neural Networks;
- support vectors in Support Vector Machines.

Model hyperparameters: external configuration variables of the model whose value can be set manually. It is difficult to know the optimal value for each hyperparameter in advance. *Tuning* a model involves finding the best configuration of hyperparameters. Examples are:

- optimizer and learning rate in Artificial Neural Networks;
- C and gamma in Support Vector Machines;
- quality measure and pruning method in Decision Trees.

Hyperparameter Tuning with GridSearch

GridSearch in practice

- Finding the right parameters can be tricky. The idea of creating a *grid* of parameters and trying all possible combinations is called *GridSearch*;
- This method is so common that Scikit-learn has this functionality built in with **GridSearchCV** (*CV* stands for *Cross Validation*);
- GridSearchCV takes a dictionary describing the parameters to be tried and the model to be trained;
- The grid parameter is defined as a dictionary where the keys are the parameters, and the values are the settings to be tested;
- GridSearchCV is a **meta-estimator** that takes an estimator such as DTC, and creates a new estimator that behaves in the same way (i.e., like a classifier);
- It is advised to add `refit=True` and set `verbose` to any number you want (*verbose* means the text output describing the process).

```
sklearn.model_selection.GridSearchCV(estimator, param_grid, , scoring=None, n_jobs=None,
                refit=True, cv=None, verbose=0, pre_dispatch='2n_jobs', error_score=nan,
                return_train_score=False)
```

```
from sklearn.model_selection import GridSearchCV
```

Hyperparameter Tuning with GridSearch

The depth of the tree and the number of leaves are obtained from the previous model:

```
print(dt_model.get_depth())
print(dt_model.get_n_leaves())
```

```
24
1189
```

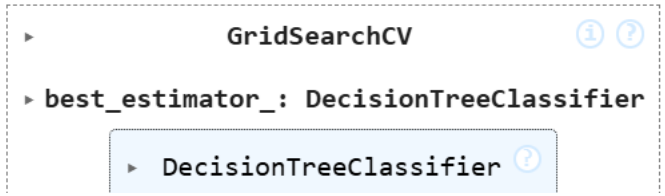
Hyperparameters to test: *criterion* and *max_depth*

```
param_grid_dt = {'criterion': ['gini', 'entropy'], 'max_depth': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]}
estimator_dt = DecisionTreeClassifier(random_state=2022)
grid_dt = GridSearchCV(estimator_dt, param_grid_dt, refit=True, verbose=2)
```

The fit:

- It runs the same loop with CV to find the best combination of parameters;
- Once it has found the best combination, it runs the fit again on all the data passed to the fit (without CV) to build a single new model using the best parameter setting.

```
grid_dt.fit(X_train, y_train)
[CV] END ....................criterion=entropy, max_depth=10; total time=   0.0s
```



Hyperparameter Tuning with GridSearch

The best estimator and hyperparameters:

```
grid_dt.best_estimator_
```

```
DecisionTreeClassifier(max_depth=6, random_state=2022)
```

Evaluate the model:

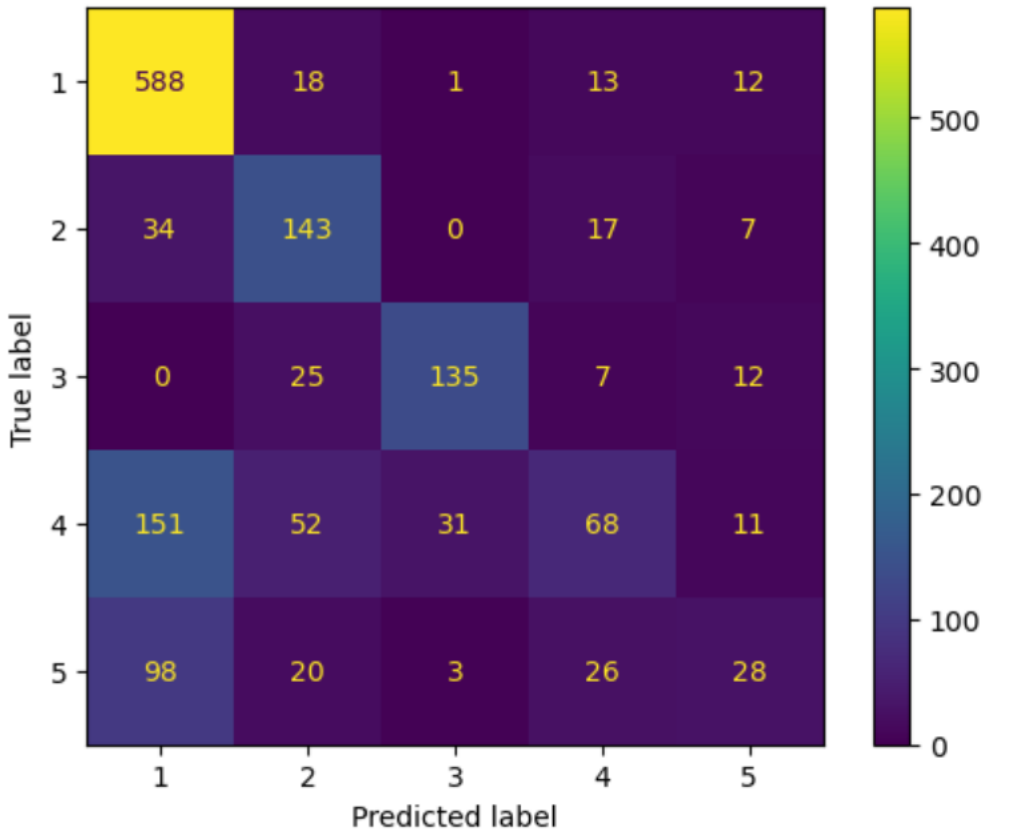
```
print(classification_report(y_test, grid_dt.predict(X_test)))
```

```
              precision    recall  f1-score   support

           1       0.68      0.93      0.78       632
           2       0.55      0.71      0.62       201
           3       0.79      0.75      0.77       179
           4       0.52      0.22      0.31       313
           5       0.40      0.16      0.23       175

    accuracy                           0.64      1500
   macro avg       0.59      0.55      0.54      1500
weighted avg       0.61      0.64      0.60      1500
```

```
ConfusionMatrixDisplay.from_predictions(y_test, grid_dt.predict(X_test))
plt.show()
```



Decision Tree Pruning

The DT is pruned by replacing an entire subtree with a leaf node. If the expected error rate of the subtree is higher than that of a single leaf, the subtree is replaced.

When to apply:

- If the node becomes very small, do not split it further;
- Minimum error (CV) pruning without early stopping is an effective technique;
- Build a full-depth tree and work backwards, using a statistical test at each stage;
- Prune an inner node and raise the subtree below it one level.

Maximum depth of the tree:

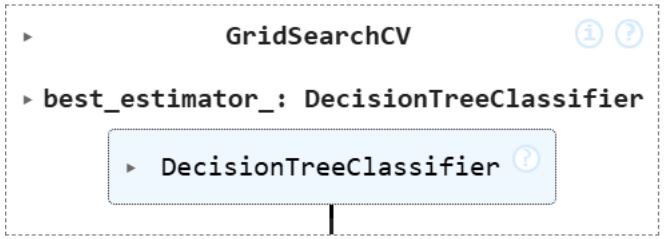
```
max_depth = dt_model.get_depth()
max_depth
```

```
24
```

Pre-pruning: *max_depth*

```
param_grid_pru = {'max_depth': [max_depth for max_depth in range(1, max_depth + 1)]}
estimator_pru = DecisionTreeClassifier(random_state=2022)
max_depth_gs = GridSearchCV(estimator_pru, param_grid_pru)
```

```
max_depth_gs.fit(X_train, y_train)
```



Decision Tree Pruning

The best hyperparameters and estimator configuration are:

```
max_depth_gs.best_params_
```

```
{'max_depth': 6}
```

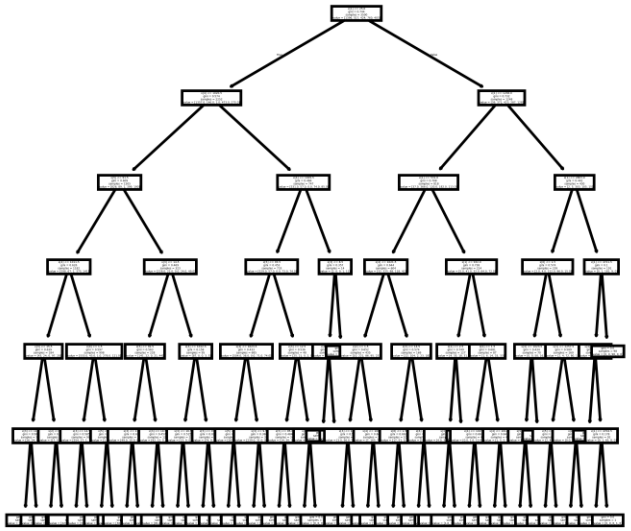
```
max_depth_tree = max_depth_gs.best_estimator_
print(max_depth_tree)
```

```
DecisionTreeClassifier(max_depth=6, random_state=2022)
```

Plot the tree:

```
fig = plt.figure(figsize=(4, 4), dpi=1000)
tree.plot_tree(max_depth_tree)
plt.show()

fig.savefig("dt_plot_md.png")
```



Decision Tree Pruning

Evaluate the model:

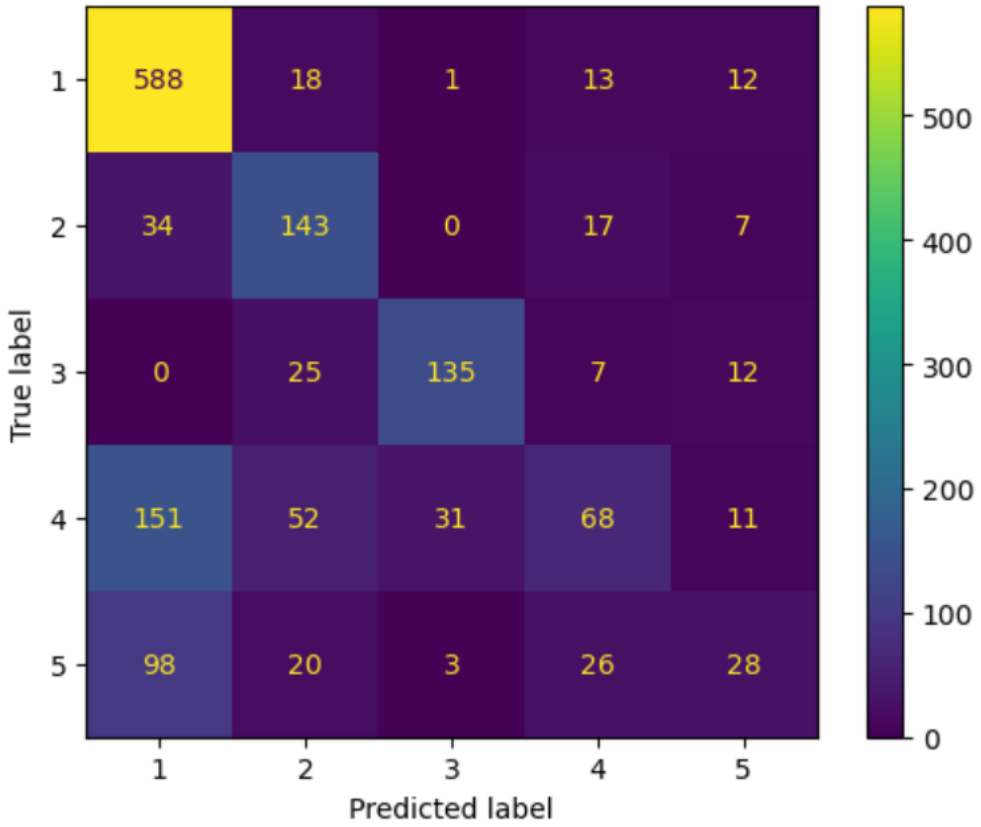
```
print(classification_report(y_test, max_depth_tree.predict(X_test)))
```

```
              precision    recall  f1-score   support

           1       0.68      0.93      0.78       632
           2       0.55      0.71      0.62       201
           3       0.79      0.75      0.77       179
           4       0.52      0.22      0.31       313
           5       0.40      0.16      0.23       175

    accuracy                           0.64      1500
   macro avg       0.59      0.55      0.54      1500
weighted avg       0.61      0.64      0.60      1500
```

```
ConfusionMatrixDisplay.from_predictions(y_test, max_depth_tree.predict(X_test))
plt.show()
```



Decision Tree Pruning

Post-pruning: *ccp_alpha*

```
ccp_alphas = dt_model.cost_complexity_pruning_path(X_train, y_train)["ccp_alphas"]
```

```
ccp_alphas
```

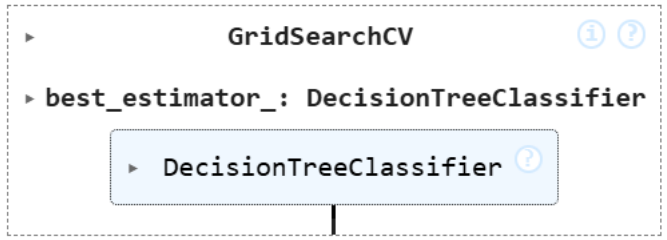
```
array([0.00000000e+00, 1.90476190e-05, 2.38095238e-05, 3.78151261e-05,
       4.76190476e-05, 6.66666667e-05, 7.14285714e-05, 7.14285714e-05,
       7.61904762e-05, 7.61904762e-05, 8.57142857e-05, 9.52380952e-05,
```

```
estimator_pru.get_params().keys()
```

```
dict_keys(['ccp_alpha', 'class_weight', 'criterion', 'max_depth', 'max_features', 'max_le
af_nodes', 'min_impurity_decrease', 'min_samples_leaf', 'min_samples_split', 'min_weight_
fraction_leaf', 'monotonic_cst', 'random_state', 'splitter'])
```

```
param_grid_pru2 = {'ccp_alpha': [alpha for alpha in ccp_alphas]}
estimator_pru2 = DecisionTreeClassifier(random_state=2022)
ccp_alpha_gs = GridSearchCV(estimator_pru2, param_grid_pru2)
```

```
ccp_alpha_gs.fit(X_train, y_train)
```



The best hyperparameters:

```
ccp_alpha_gs.best_params_
```

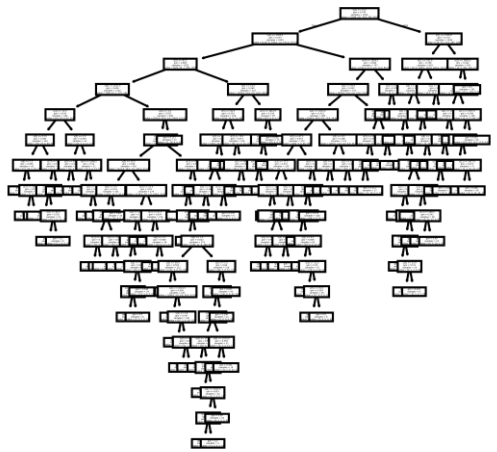
```
{'ccp_alpha': 0.0008316875137151618}
```

```
best_ccp_alpha_tree = ccp_alpha_gs.best_estimator_
print(best_ccp_alpha_tree)
```

```
DecisionTreeClassifier(ccp_alpha=0.0008316875137151618, random_state=2022)
```

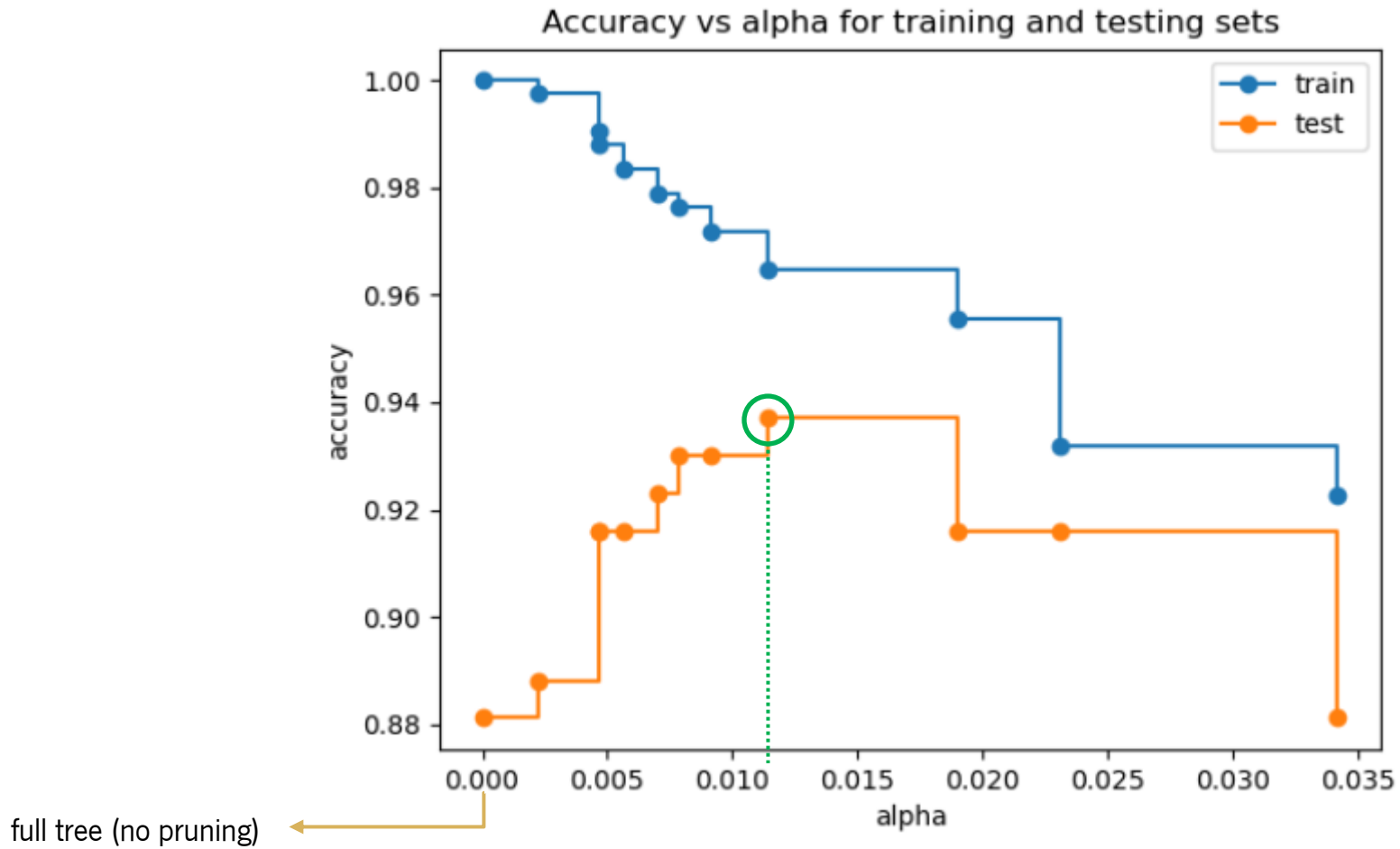
```
fig = plt.figure(figsize=(4, 4), dpi=1000)
tree.plot_tree(best_ccp_alpha_tree)
plt.show()

fig.savefig("dt_plot_alpha.png")
```



Decision Tree Pruning

The following graph presents an example of how the value of *ccp_alpha* can impact the performance of the model.



Decision Tree Pruning

Evaluate the model:

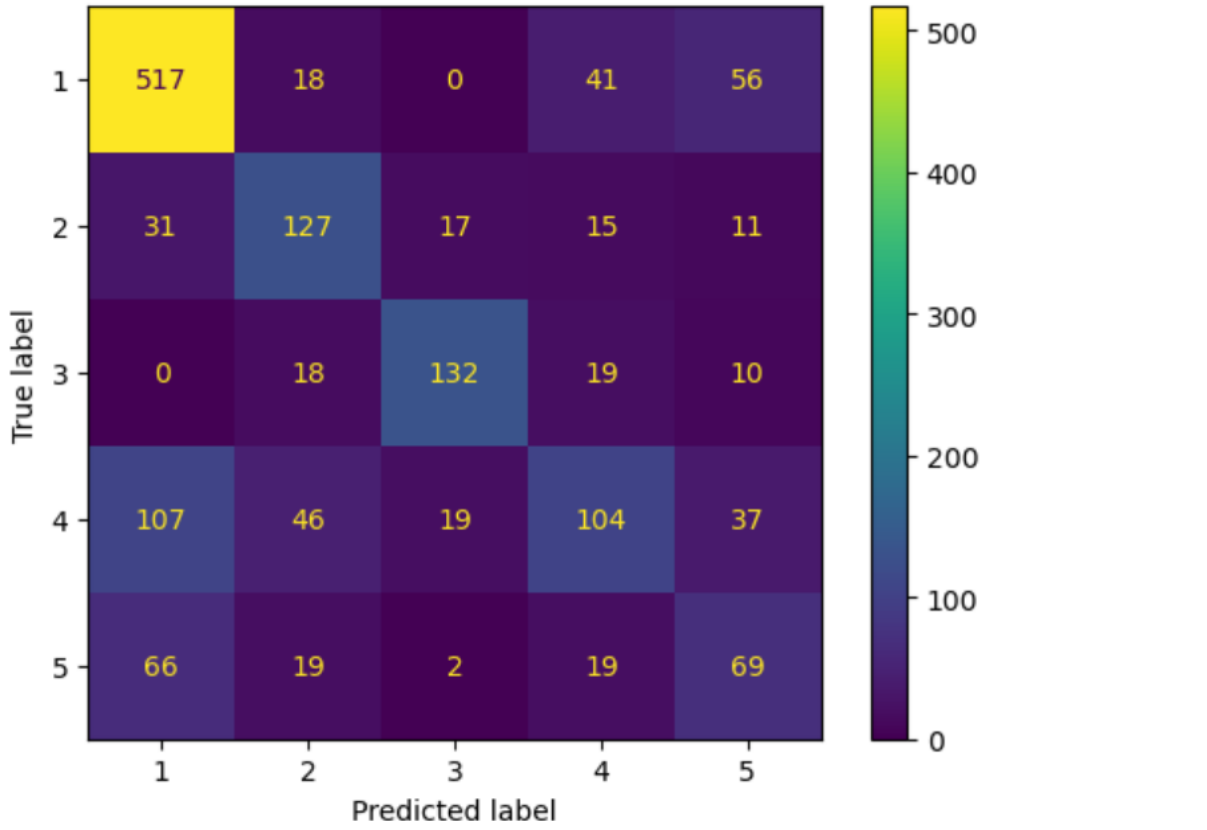
```
print(classification_report(y_test, best_ccp_alpha_tree.predict(X_test)))
```

```
              precision    recall  f1-score   support

           1       0.72      0.82      0.76       632
           2       0.56      0.63      0.59       201
           3       0.78      0.74      0.76       179
           4       0.53      0.33      0.41       313
           5       0.38      0.39      0.39       175

    accuracy                           0.63      1500
   macro avg       0.59      0.58      0.58      1500
weighted avg       0.62      0.63      0.62      1500
```

```
ConfusionMatrixDisplay.from_predictions(y_test, best_ccp_alpha_tree.predict(X_test))
plt.show()
```



Hands On